

Strategic coexistence theory for evolutionary game theory

Sang Woo Park^{1,*}

¹ School of Biological Sciences, Seoul National University, Seoul, Korea

*Corresponding author: sangwoopark@snu.ac.kr

Abstract

Introduction

Evolutionary game theory (EGT) provides a powerful framework for predicting which competing strategies are evolutionarily stable. Because the success of each strategy depends on the frequency of other strategies, even simple two-strategy games can yield heterogeneous outcomes, including the dominance of one strategy (prisoner's dilemma and harmony games), coexistence (hawk–dove games), and bistability (stag-hunt games). These differences are typically understood through payoff inequalities, invasion criteria, or stability analysis, but Tarnita and Traulsen (2025) recently argued that an ecological perspective on EGT may yield new, ecological insights. In particular, recasting game dynamics as a coexistence problem immediately raises two important questions: when two strategies coexist, what mechanism stabilizes it? And when coexistence fails, is it because stabilizing effects are absent, or because fitness differences are too large?

Modern coexistence theory (MCT) from community ecology presents a particularly useful framework for answering these questions. MCT predicts that coexistence of two competing entities, be they species or strategies, requires stabilizing niche differences to be greater than average fitness differences. In community ecology, stabilizing niche differences represent differences in factors that limit the growth of competing species, causing each species to limit itself more than it limits its competitor. In EGT, a strategy's niche can be interpreted as strategic environments generated by the composition of competing strategies and their interactions. Thus, stabilizing niche differences arise when strategies are favored by different strategic environments, such that the spread of each strategy creates conditions that limit its own relative growth and favor the recovery of its competitor. In contrast, fitness differences represent differences in the ability of each strategy to compete under the limiting strategic environments created by its competitor. This differs from payoff differences in EGT, which depend on the underlying strategy frequency and therefore combine both stabilizing effects caused by changes in the strategic environment and residual fitness differences that favor one strategy over another.

So far, MCT has been primarily utilized for understanding species coexistence, but recent work showed that MCT can be extended beyond species competition, especially for studying pathogen strain competition, allowing one to directly quantify

niche and fitness differences of competing entities, be they species or strains. Inspired by this effort, we present a Strategic Coexistence Theory (SCT) for predicting coexistence and exclusion of competing strategies in EGT by quantifying strategic niche and fitness differences. We begin by introducing SCT, which can be applied to any generic system of EGT described by replicator equations.

Strategic coexistence theory

We begin with a classic replicator equation, which allows us to model changes in the frequency of competing strategies based on their payoffs:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (1)$$

where x_i represents the frequency of strategy i , $\pi_i(\mathbf{x})$ represents the payoff of strategy i for a given strategy distribution $\mathbf{x}$, and $\bar{\pi}(\mathbf{x}) = \sum_i \pi_i(\mathbf{x})x_i$ represents the average payoff. Since strategy frequencies must sum to 1 (i.e., $\sum_i x_i = 1$), a single-strategy equilibrium dominated by strategy i can be described by a unit vector $\mathbf{e}_i$, where all elements are equal to zero except for the i -th element. We use these single-strategy equilibria to define niche and fitness differences between two competing strategies i and j . However, because payoffs in many games can be negative, payoff ratios are difficult to interpret. Thus, we work on a positive multiplicative scale by exponentiating the payoff terms:

$$W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x})). \quad (2)$$

For convenience, we refer to $W_i(\mathbf{x})$ as the multiplicative payoff of strategy i .

First, we define the niche overlap ρ between strategies i and j as the ratio between the geometric mean of each strategy's multiplicative payoff in the strategic environment that it creates when common, $\sqrt{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}$, and the geometric mean of each strategy's multiplicative payoff in the environment generated by its competitor, $\sqrt{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}$. The corresponding niche difference is then given by

$$1 - \rho = 1 - \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (3)$$

When $1 - \rho > 0$, each strategy performs better, on average, in the environment generated by its competitor than in the environment generated by itself. In other words, intraspecific competition is stronger than interspecific competition.

Second, we define the fitness difference as the ratio between the geometric means of the multiplicative payoff of each strategy across the two single-strategy environments:

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (4)$$

69 In other words, strategy j has an average competitive advantage over strategy i
70 ($\kappa_j/\kappa_i > 1$) when it performs better, on average, across the environments generated
71 by itself and its competitor. Then, analogous to MCT, mutual invasion requires the
72 fitness ratio to fall within the bounds set by niche overlap:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (5)$$

73 This decomposition allows us to understand how different mechanisms contribute to
74 the coexistence or exclusion of competing strategies.

75 Two strategy games

76 As an illustrative example, we begin with classic two-strategy games, which can be
77 described by the following payoff matrix (Figure 1A):

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (6)$$

78 Here, the first and second rows represent strategies 1 and 2, respectively, and the
79 columns represent the strategy of the interacting opponent. Writing x_1 to denote
80 the frequency of strategy 1, the expected payoffs are

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (7)$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (8)$$

81 Thus, the dynamics can be written as

$$\frac{dx_1}{dt} = x_1(1 - x_1) [(R - T)x_1 + (S - P)(1 - x_1)]. \quad (9)$$

82 The signs of $T - R$ and $S - P$ determine whether each strategy can invade the
83 other when rare. Specifically, strategy 2 can invade strategy 1 when $T > R$, whereas
84 strategy 1 can invade strategy 2 when $S > P$. These two invasion conditions partition
85 the game space into four regimes (Figure 1B): prisoner's dilemma ($T > R$, $P >$
86 S), hawk-dove game ($T > R$, $S > P$), stag-hunt game ($R > T$, $P > S$), and
87 harmony game ($R > T$, $S > P$). In standard EGT, these outcomes are obtained by
88 analyzing the stability of equilibrium conditions. For simple games, this analysis is
89 straightforward.

90 A coexistence-theoretic approach recovers the same classification but provides
91 a different interpretation (Figure 1C). For a two-strategy game determined by this
92 payoff matrix, the niche difference and fitness difference are given by

$$1 - \rho = 1 - \exp\left(\frac{R + P - S - T}{2}\right), \quad (10)$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (11)$$

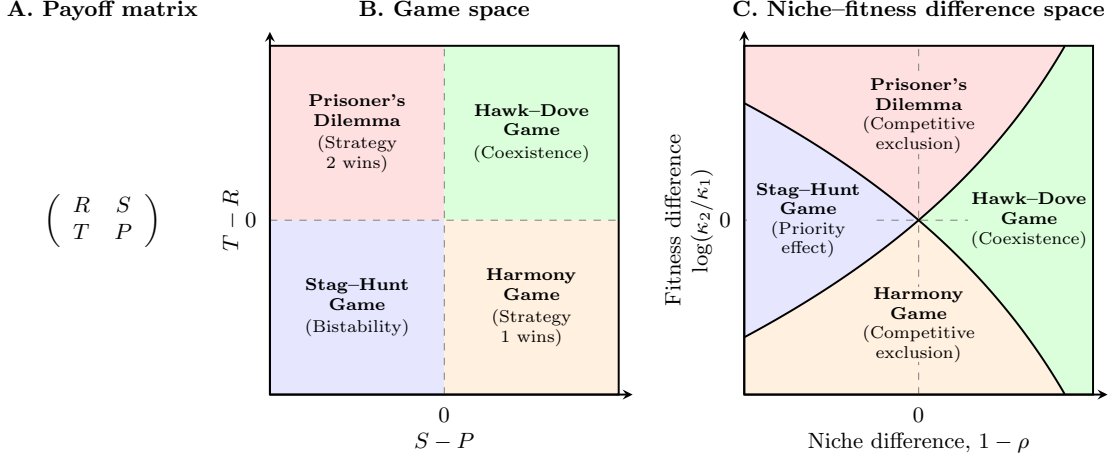


Figure 1: **Coexistence-theoretic interpretation of classic two-strategy games.** (A) General payoff matrix for a two-strategy game, where rows represent the focal strategy and columns represent the strategy of the interacting opponent. (B) Classification of four classic games based on the payoff differences $T - R$ and $S - P$. Prisoner's dilemma and harmony games lead to the dominance of one strategy; hawk-dove games allow coexistence; and stag-hunt games result in bistability. (C) Mapping of the same four games onto niche-fitness difference space. Hawk-dove games correspond to the coexistence regime, where stabilizing niche differences overcome fitness differences. Prisoner's dilemma and harmony games correspond to competitive-exclusion regimes, whereas stag-hunt games correspond to the priority-effect regime.

93 As shown above, mutual invasion requires $\rho < \kappa_2/\kappa_1 < 1/\rho$, which is equivalent to
 94 $T > R$ and $S > P$. This corresponds to the hawk-dove game, in which each strategy
 95 can invade when rare and the two strategies coexist. The remaining games also
 96 have natural interpretations in niche-fitness difference space. Prisoner's dilemma
 97 and harmony games represent competitive exclusion regimes, where one strategy
 98 has a fitness advantage that exceeds niche differences. In contrast, stag-hunt games
 99 represent a priority-effect regime, where neither strategy can invade the other and
 100 the outcome depends on initial conditions. Therefore, coexistence theory not only
 101 allows us to recover the classic results from two-strategy games but also provides new
 102 interpretations for these results in terms of niche and fitness differences. Building on
 103 this, we next apply this framework to microbial cooperation, where nonlinear benefits
 104 and environmental context can alter the coexistence of cooperators and defectors.

105 Microbial cooperation and public goods

106 Here, we consider a simple, phenomenological model of competition between cooper-
 107 ating and cheating yeast in a cooperative sucrose-metabolism system, where invertase
 108 production by cooperators allows cheaters to utilize glucose, a public good released
 109 through sucrose hydrolysis (Gore et al., 2009). Specifically, invertase production by
 110 cooperators incurs a cost c and generates a total benefit of 1 that is captured pri-
 111 vately with efficiency ϵ . Assuming a linear relationship between glucose availability
 112 and microbial growth, the payoffs for cooperators π_C and cheaters (defectors) π_D can
 113 then be written as

$$\pi_C(f) = \epsilon + f(1 - \epsilon) - c, \quad (12)$$

$$\pi_D(f) = f(1 - \epsilon), \quad (13)$$

114 where the use of glucose depends on the fraction of cooperators f . The dynamics of
 115 cooperators and defectors are described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\pi_C(f) - \pi_D(f)] \quad (14)$$

$$= f(1 - f)(\epsilon - c) \quad (15)$$

116 For this linear growth model, the payoff difference is fixed to $\epsilon - c$ independent of
 117 f (Figure 2A). Therefore, cooperation is favored when efficiency is greater than the
 118 cost of cooperation, $\epsilon > c$, whereas defection is favored when the cost is greater than
 119 efficiency, $c > \epsilon$ (Gore et al., 2009).

120 Our framework allows us to reinterpret this result from the perspective of coexis-
 121 tence theory. In particular, the two strategies have no niche difference, corresponding
 122 to complete niche overlap, $\rho = 1$. In contrast, the fitness difference is determined by
 123 the balance between private efficiency and the cost of cooperation:

$$\frac{\kappa_D}{\kappa_C} = \exp(c - \epsilon). \quad (16)$$

124 Therefore, this model always predicts competitive exclusion, except in the neutral
 125 case $c = \epsilon$. The outcome is determined entirely by the fitness difference between
 126 cooperators and defectors, rather than by stabilizing niche differences.

127 In real biological systems, the effect of glucose on microbial growth can follow a
 128 nonlinear relationship:

$$\pi_C(f) = [\epsilon + f(1 - \epsilon)]^\alpha - c, \quad (17)$$

$$\pi_D(f) = [f(1 - \epsilon)]^\alpha, \quad (18)$$

129 where $\alpha = 0.15$ reflects the experimental relationship between glucose and microbial
 130 growth (Gore et al., 2009). Then, the dynamics of cooperators and defectors are
 131 described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\{\epsilon + f(1 - \epsilon)\}^\alpha - c - f(1 - \epsilon)^\alpha]. \quad (19)$$

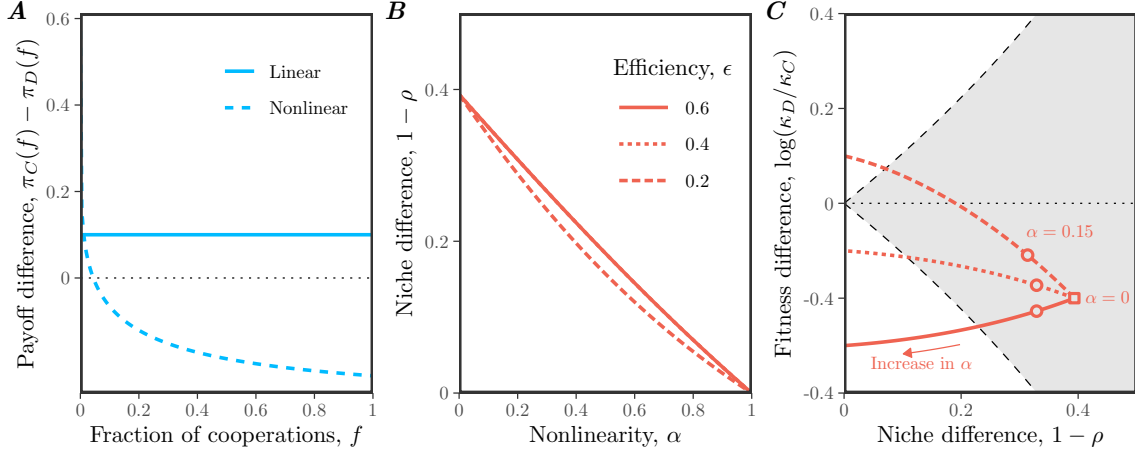


Figure 2: Caption. (A) Payoff differences for linear and nonlinear models of microbial growth. In panel A, we assumed $\epsilon = 0.4$, $c = 0.2$, and $\alpha = 0.15$ for illustrative purposes. (B) Effect of nonlinearity α on niche difference $1 - \rho$. Each line assumes a different value of efficiency ϵ . (C) Effect of nonlinearity α on both niche difference $1 - \rho$ and fitness difference $\log(\kappa_D/\kappa_C)$. The gray shaded area represents the coexistence regime. The square represents the condition when $\alpha = 0$. The circles represent the condition when $\alpha = 0.15$, which correspond to the experimental relationship between glucose and microbial growth (Gore et al., 2009). In panels B and C, we assumed $c = 0.2$, while allowing ϵ and α to vary.

Under this nonlinear assumption, the payoff difference decreases with the cooperator fraction f , which can promote the coexistence of cooperators and defectors (Figure 2A): cooperators have a competitive advantage when rare, whereas defectors have a competitive advantage when cooperators are abundant.

From a coexistence-theoretic perspective, this means that nonlinear growth can generate sufficient stabilizing niche difference to overcome the fitness difference. In particular, we can show that the niche and fitness differences for this model are given by

$$1 - \rho = 1 - \exp\left(\frac{1 - \epsilon^\alpha - (1 - \epsilon)^\alpha}{2}\right), \quad (20)$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(c + \frac{(1 - \epsilon)^\alpha - 1 - \epsilon^\alpha}{2}\right). \quad (21)$$

These expressions clarify the separate roles of cost, efficiency, and nonlinearity. For example, the cost of cooperation does not affect niche difference. Instead, niche difference is generated by the concave relationship between glucose availability and microbial growth, which increases from 0 to $1 - \exp(-1/2)$ as α goes from 1 to 0 (Figure 2B,C). In contrast, as $\alpha \rightarrow 0$, the fitness difference approaches

$$\log(\kappa_D/\kappa_C) \rightarrow c - \frac{1}{2}, \quad (22)$$

145 which no longer depends on the private capture efficiency ϵ (Figure 2C). Thus, strong
146 concavity not only generates stabilizing niche differences, but also acts as an equal-
147 izing mechanism (Figure 2C).

148 **Supplementary Text**

149 **References**

- 150 Gore, J., H. Youk, and A. Van Oudenaarden (2009). Snowdrift game dynamics and
151 facultative cheating in yeast. *Nature* *459*(7244), 253–256.
- 152 Tarnita, C. E. and A. Traulsen (2025). Reconciling ecology and evolutionary game
153 theory or “When not to think cooperation”. *Proceedings of the National Academy*
154 *of Sciences* *122*(14), e2413847122.